SANELLA PALANGIC

IT Systems Administration • AI Automation • Data Science

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Portfolio: **sanellapalangic2.github.io**



PROFESSIONAL SUMMARY

Data Science M.S. candidate and IT/AI professional with 4+ years of experience administering systems and supporting users, devices, and Microsoft 365 in a professional office. Automates business workflows with AI (Anthropic Claude), supports network security, cloud backup, accounting, and CRM systems, and builds and deploys Python, SQL, and machine learning applications, including LLM and NLP solutions.

EDUCATION

Master of Science in Data Science | *Montclair State University, Montclair, NJ* — **Expected May 2028**
GPA: 3.5+ / 4.0 • Partial Academic Scholarship

B.S. in Computer Science, Minor in Mathematics | *Fairleigh Dickinson University, Teaneck, NJ* — **May 2026**
GPA: 3.5+ / 4.0 • Dean's List • Partial Academic Scholarship

PROFESSIONAL EXPERIENCE

E-PARALEGALS LLC — **Oradell, NJ**

IT/AI Specialist *(Part-Time, 15 hrs/week)* — **Jan 2024 – Present**

- Automate composition of complex legal filings and recurring tasks with Claude AI, Python, and parallelized Google Colab scripts
- Maintain network security across the office network and connected devices, safeguarding confidential firm data
- Ensure reliable cloud backup of firm data via Dropbox, keeping business files protected and recoverable
- Build and maintain Airtable bases to track projects, assignments, and workflow status
- Synchronize accounting data with QuickBooks Online to keep financial records accurate and consistent across systems
- Assist with FluentCRM email marketing campaigns and the corresponding campaign tracking in Amazon Web Services (AWS)

IT Support and Systems Administrator — **Jan 2022 – Present**

- Deliver user support and troubleshooting for hardware, software, and account-related issues in a professional office environment
- Install, image, and configure laptops and desktops, ensuring proper software setup and security settings
- Support Microsoft 365 applications (Outlook, Teams, OneDrive), resolving access and synchronization issues
- Maintain and troubleshoot network connectivity, including Wi-Fi, VPN, and printers
- Document common technical issues and step-by-step solutions, contributing to the internal IT knowledge base

TECHNICAL PROJECTS

TaskFlow AI: AI Operations Command Center | *Python, Flask, SQLite, Anthropic Claude API, Pydantic, JavaScript* — **Live Demo** | **GitHub**

- Built and deployed a full-stack LLM operations platform with human-in-the-loop design: the model only suggests, while business rules and a human approval queue make every change, logged in an append-only audit trail
- Engineered LLM guardrails: strict JSON-schema outputs validated with Pydantic, a tool-calling copilot limited to 10 read-only SQL queries, and grounding checks that withhold answers not backed by retrieved data
- Built an evaluation lab benchmarking the model on 40 labeled test cases, improving classification accuracy from 69% to 82% across runs at $0.0066 per analysis, with regression tracking

NYC 311 Service Request Analysis | *Python, PostgreSQL, SQL, pandas, scikit-learn, lifelines, Streamlit* — **Live Demo** | **GitHub**

- Built a pipeline loading 3.66M NYC 311 requests into PostgreSQL and trained a leakage-free gradient boosting model on a time-based split predicting 7-day closure (0.956 ROC AUC vs. a 0.50 baseline)
- Applied survival analysis and statistical testing; deployed a Streamlit dashboard validated against SQL

IT Support Ticket Classifier | *Python, scikit-learn, pandas, NumPy, Streamlit* — **Live Demo** | **GitHub**

- Built an NLP classifier (TF-IDF, logistic regression) routing IT tickets across 8 categories, trained on 47,837 labeled tickets and reaching 0.865 macro F1 on a 6,959-ticket held-out test set
- Prevented leakage with near-duplicate clustering and group-aware splits; deployed on Render with 24 pytest tests

SKILLS & LANGUAGES

AI & LLM Engineering	Anthropic Claude API, structured outputs, tool calling, prompt engineering, human-in-the-loop, LLM evals
Machine Learning	scikit-learn, NLP, TF-IDF, logistic regression, gradient boosting, cross-validation, model evaluation
Programming & Frameworks	Python, SQL, JavaScript, React, Flask, Streamlit, Pydantic, REST APIs, HTML/CSS
Data & Analytics	PostgreSQL, SQLite, SQL Server, pandas, NumPy, Power BI, Google Colab
Cloud & DevOps	Microsoft Azure, Amazon Web Services (AWS), Microsoft Entra ID, Render, Git, GitHub, pytest
Business Systems	Microsoft 365, Dropbox, Airtable, QuickBooks Online, FluentCRM
Languages	English (Native/Fluent) • Spanish (Conversational Proficiency)